

# Multi-task Medical Image Analysis with Diffusion-Based Synthetic Augmentation

## Clinical Performance Report

|              |                                 |             |                         |
|--------------|---------------------------------|-------------|-------------------------|
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| Supervisor   | Dr. Muhammad Farooq             | Course      | Medical Image Computing |
| Dataset      | Breast Ultrasound Images (BUSI) | Report Date | 2026-06-13 14:32        |

### 1. Segmentation Results (U-Net)

| Metric      | Baseline (Real Only) | Augmented (Best Ratio) | Target |
|-------------|----------------------|------------------------|--------|
| Dice Score  | —                    | —                      | > 0.75 |
| IoU         | —                    | —                      | > 0.60 |
| Sensitivity | —                    | —                      | > 0.80 |
| Specificity | —                    | —                      | > 0.85 |

### 2. Classification Results (ResNet-50)

| Metric         | Baseline (Real Only) | Augmented (Best Ratio) | Target |
|----------------|----------------------|------------------------|--------|
| AUC (OvR)      | —                    | —                      | > 0.85 |
| Accuracy       | —                    | —                      | > 0.80 |
| F1 – Normal    | —                    | —                      | > 0.78 |
| F1 – Benign    | —                    | —                      | > 0.78 |
| F1 – Malignant | —                    | —                      | > 0.78 |

### 3. Synthetic Image Quality (Diffusion Model)

| Metric           | Value | Interpretation                                |
|------------------|-------|-----------------------------------------------|
| FID Score        | —     | Lower is better (real ↔ synthetic similarity) |
| Images Generated | —     | Total synthetic images across all classes     |